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Real-time and file-based audio data processing

Abstract

This application is directed to dynamically transferring audio data in real time or in batch. A microphone of a first electronic device captures audio signals. The audio signals are sampled to provide a first sequence of audio data samples and a second sequence of audio data samples that follows the first sequence. The second electronic device receives the first and second sequences of audio data samples, and processes the first sequence of audio data samples into audio data packets according to a real time data processing mode. The second electronic device determines that the second electronic device cannot support processing of audio data samples in the real time data processing mode, caches the second sequence of audio data samples in a buffer, and generates a data file including the second sequence of audio data samples in a batch data processing mode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/086,953, filed Oct. 2, 2020, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) This application relates generally to audio data transfer and processing, including but not limited to methods and systems for providing real time and file-based audio data processing to facilitate data transfer and speech recognition in electronic devices.

BACKGROUND

(2) Electronic devices having microphones are widely used as assistant devices to collect voice inputs from users and initiate different voice-activated functions according to the voice inputs. For example, many remote control devices coupled to digital television devices are configured to integrate microphones. Voice inputs of these remote control devices are streamed to and at least partially processed by the digital television devices. The digital television devices may submit the voice inputs (pre-processed or not) to a remote server system for additional audio processing. User requests are extracted from the voice inputs for initiating the voice-activated functions as a result of audio processing at the television devices and/or remote server system. Any deficiency in communication, caching and processing capabilities of the television devices can cause a delay in the initiation of the voice-activated functions. This often happens to low-cost television devices that have limited capabilities. It would be beneficial to have a more efficient data processing and transfer mechanism than the current practice to compensate for the deficiencies in these devices' communication, caching or processing capabilities.

SUMMARY

(3) This application is directed to processing and transfer audio data received from an electronic device having a microphone (e.g., a remote control device, an assistant device). The electronic device is coupled to another electronic device (e.g., a television device) that has audio processing capabilities or is coupled to a server having audio processing capabilities. These two electronic devices are coupled via a communication channel. The audio data is transferred in real time via the communication channel, and processed in real time or in batch by the receiving electronic device depending on whether communication, computation and storage of the receiving electronic device can support processing of audio data samples in real time. Real time audio data processing is thereby supplemented by the batch audio data processing, particularly in some electronic devices that do not always have sufficient resources to communicate, cache or process the audio data in real time.

(4) Specifically, in one aspect, a method is implemented to process audio data, e.g., switch from a real time data processing mode to a batch data processing mode. The method includes receiving, from a first electronic device (e.g., a remote control device) and by a second electronic device (e.g.,

a television device), a first sequence of audio data samples and a second sequence of audio data samples. The second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device. The method further includes processing by the second electronic device the first sequence of audio data samples according to a real time data processing mode and determining that the second electronic device cannot support processing of audio data samples in the real time data processing mode. The method further includes in accordance with the determination that the second electronic device cannot support processing of audio data samples in the real time data processing mode, caching, in a buffer of the second electronic device, the second sequence of audio data samples and generating a data file including the second sequence of audio data samples in a batch data processing mode.

(5) Alternatively, in another aspect, a method is implemented to process audio data, e.g., switch from a batch data processing mode to a real time data processing mode. The method includes receiving, from a first electronic device (e.g., a remote control device) and by a second electronic device (e.g., a television device), a first sequence of audio data samples and a second sequence of audio data samples. The second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device. The method further includes processing by the second electronic device the first sequence of audio data samples according to a batch data processing mode, which further includes caching, in a buffer of a second electronic device, the first sequence of audio data samples, and generating a data file including the first sequence of audio data samples. The method further includes determining that the second electronic device can support processing of audio data samples in a real time data processing mode. The method further includes in accordance with the determination that the second electronic device can support processing of audio data samples in the real time data processing mode, processing by the second electronic device the second sequence of audio data samples according to the real time data processing mode.

(6) A non-transitory computer-readable medium has instructions stored thereon, which when executed by one or more processors cause the processors to perform the method of any of the above methods. An electronic device includes one or more processors and memory having instructions stored thereon, which when executed by the one or more processors cause the processors to perform the method of any of the above methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the various described implementations, reference should be made to the Description of Implementations below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

(2) FIG. 1 is an example media environment in which a network-connected TV device, a remote control device, and a server system interact with each other via one or more communication networks, in accordance with some implementations.

(3) FIG. 2 is an example audio data transfer path between a first electronic device and a second electronic device, in accordance with some implementations.

(4) FIG. 3 is a schematic diagram illustrating an example audio data processing process switching from a real time data processing mode to a batch data processing mode, in accordance with some implementations.

(5) FIG. 4 is a schematic diagram illustrating an example audio data processing process switching from a batch data processing mode to a real time data processing mode, in accordance with some implementations.

(6) FIG. 5 is a schematic diagram illustrating an example voice assistant process initiated by a user

action or a voice input, in accordance with some implementations.

(7) FIG. 6 illustrates an example remote control device configured to transfer audio data to a television device, in accordance with some implementations.

(8) FIG. 7 is a flow chart of a method for processing audio data dynamically in two audio data processing modes, in accordance with some implementations.

(9) FIG. 8 is a flow chart of another method for processing audio data dynamically in two audio data processing modes, in accordance with some implementations.

(10) Like reference numerals refer to corresponding parts throughout the several views of the drawings.

DESCRIPTION OF THE IMPLEMENTATIONS

(11) Many electronic devices (e.g., a remote control device, voice-activated display or speaker devices) include microphones used to collect voice inputs from an environment where the electronic devices are disposed. Such an electronic device may be configured to collect the voice inputs automatically upon detecting hot words in an audio signal or in response to a user press on a dedicated assistant button of the electronic device. Upon receiving a voice input, the electronic device communicates the voice input to a remote server system (e.g., an assistant server) via one or more communication network, and the remote server system recognizes a user request in the voice input and responds to the user request. In an example, the electronic device includes a remote control device, and the voice input is initiated to control a network-connected television (TV) device coupled to the remote control device. The remote control device sends the voice input to the remote server system via the TV device, and the voice input is processed at the TV device before the TV device sends the voice input to the remote server system. During this process of processing the voice input, the TV device uses its communication, computation and storage capabilities to bridge the remote control device and the remote server system.

(12) Audio data delivered to an audio manager of the TV device may be different from what is collected by a microphone of the remote control device. This happens due to various factors, e.g., loss and delay of data packets via a communication channel that couples the remote control and TV devices, processor load of the TV device. In various implementations of this application, a combination of real time data processing and file-based batch data processing is implemented at a second electronic device (e.g., the TV device) to process audio data collected by a microphone of a first electronic device (e.g., the remote control device). In some implementations, real-time data processing has a priority over file-based batch data processing. When it is determined that at least one of communication, computation and storage capabilities of the second electronic device cannot support processing of audio data samples in real time, subsequent data samples are cached and organized into a data file at the second electronic device (e.g., processed by an audio data processing module distinct from the audio manager). Alternatively, when it is determined that communication, computation and storage capabilities of the second electronic device can support processing of audio data samples in real time, subsequent data samples are processed into data packets by the second electronic device in real time (e.g., processed by the audio manager, which is part of an operating system of the second electronic device). Such a controlled audio data transfer process streamlines data processing and communication at the second electronic device and improves performance, audio quality, and user experience associated with voice inputs that initiate user interactions with the electronic devices.

(13) Reference will now be made in detail to implementations, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described implementations.

However, it will be apparent to one of ordinary skill in the art that the various described implementations may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the implementations.

(14) FIG. 1 is an example media environment **100** in which a network-connected TV device **102**, a remote control device **104**, and a server system **106** interact with each other via one or more communication networks **180**, in accordance with some implementations. The media environment **100** corresponds to a virtual user domain created and hosted by the server system **106**, and the virtual user domain includes a plurality of user accounts. For each user account, the server system **106** is coupled to a content source **110** and one or more media devices **102** and **116-126**, and is configured to stream media content provided by the content source **110** for review by a user via the respective user account. Optionally, the content source **110** includes one or more of: ads sources, electronic program guide (EPG) sources, and media content sources.

(15) Specifically, the one or more media devices associated with the user and the user account are disposed in the media environment **100** to provide the user with media content that is stored at and streamed from the content source **110**. The content source **110** is optionally a third-party media content source or an internal media source hosted by the server system **106**. In some implementations, the one or more media devices include the network-connected TV device **102** which directly streams the media content from the remote content source or integrates an embedded casting unit configured to stream the media content for display to its audience. The network-connected TV device **102** is communicatively coupled to a dedicated remote control device **104** and/or an electronic device having a remote control application (e.g., a mobile phone **122**, a tablet computer **124**, a laptop computer **126**, an assistant device **138**). The dedicated remote control device **104** may be disposed in proximity to the TV device **102** and configured to communicate with the TV device **102** with digitally-coded pulses of infrared signals. Alternatively, in some situations, the dedicated remote control device **104** or electronic device having the remote control application is configured to communicate with the TV device **102** via the communication networks **180** (i.e., via a short-range communication link, a local area network and/or a wide area network), and does not have to be physically in proximity to the TV device **102**.

(16) The network-connected TV device **102** includes one or more processors and memory storing instructions for execution by the one or more processors. The instructions stored on the network-connected TV device **102** include one or more of: a unified TV application, a local content casting application, a remote control application, an assistant application, and one or more media play applications associated with the content source **110**. These applications are user applications distinct from an operating system of the TV device **102**, and are optionally linked to the user account in the virtual user domain of the media environment **100**. Additionally, the network-connected TV device **102** includes an audio manager (e.g., **234** in FIG. 2) integrated in its operating system to process audio data on a data packet level.

(17) Alternatively, in some implementations, the media devices disposed in the media environment **100** include a display device **116** that directly outputs the media content to an audience and a casting device **118** that is coupled to and configured to stream the media content to the display devices **116**. Examples of the display device **116** include, but are not limited to, television (TV) display devices and music players. Examples of the casting device **118** include, but are not limited to, set-top boxes (STBs), DVD players and TV boxes. In this example shown in FIG. 1, the display device **116** includes a TV display that is hard wired to a DVD player or a set-top box **118**. In contrast, in some implementations, the media devices disposed in the media environment **100** include a computer screen **120A** that outputs the media content to audience and a desktop computer **120B** that streams the media content to the computer screen **120A**. In some implementations, the media devices disposed in the media environment **100** includes a mobile device, e.g., a mobile phone **122**, a tablet computer **124** and a laptop computer **126**. Each of the media devices **118-126** includes one or more media play applications configured to receive and play media content items provided by the content source **110** or internal media source associated with the server system **106**.

(18) The server system **106** includes a unified media platform (I-JWP) **128** configured to manage media content recommendation and streaming for the one or more media devices in the media

environment **100**. Media content recommendations generated by the I-AAP **128** are presented on the network-connected TV device **102** via a server-side TV application **134**, and the server-side TV application **134** enables display of media content on the unified TV application on the TV device **102** in response to a user selection from the media content recommendations. Moreover, the UNIP **128** may also function as a centralized media content management module configured to provide the media content recommendations to other media devices **118-126** in addition to the TV device **102**. In some implementations, activity data associated with each user account is collected from the TV application **134** and casting service module **136**, and used to personalize the media content recommendations provided to a user of the user account.

(19) In some implementations, in addition to one or more of the media devices **102**, **104**, and **116-126**, a user account of the virtual user domain hosted by the server system **106** is also associated with one or more devices of other types, e.g., network-connected assistant devices **138** that are installed in the media environment **100**. Examples of the assistant devices **138** include a speaker assistant device **142** and a display assistant device **144**. The speaker assistant device **142** is capable of collecting audio inputs, recognizing user commands from the audio inputs, and implementing operations (e.g., playing music, answering questions) in response to the user commands. The display assistant device **144** is capable of collecting audio and/or video inputs, recognizing user commands from the audio and/or video inputs, and implementing operations (e.g., playing music, presenting an image or video clip, answering questions) in response to the user commands. Each of the assistant devices **138** is optionally managed by a dedicated device application or a generic user application (e.g., a web browser), and linked to the user account in the virtual domain in conjunction with the unified TV application of the network-connected TV device **102**.

(20) Further, in some implementations, the server system **106** includes an assistant module **140** that is optionally powered by artificial intelligence. The assistant module **140** is configured to recognize user requests from voice inputs collected by a microphone and initiate operations to search the Internet, schedule events and alarms, adjust hardware settings, present public or private information, play media content items, have a two-way conversation with a user, purchase products, transfer money, or the like. The microphone is integrated in any of the media devices **102**, **104**, and **116-126** and the assistant devices **138** disposed in the media environment **100**. In some implementations, the assistant module **140** is coupled to a speech recognition module **160** that is configured to process the voice inputs collected by the microphone and identify user requests from the voice inputs, e.g., using natural language processing (NLP) algorithms.

(21) In some implementations, the server system **106** includes a device and application registry **150** configured for storing information of one or more user accounts managed by the server system **106** and information of user devices and applications associated with each of the one or more user accounts. For example, the device and application registry **150** stores information of the network-connected TV device **102**, remote control device **104**, media devices **116-126**, assistant devices **138** and information of the corresponding unified TV application, remote control application, media play applications, and dedicated device applications associated with the assistant devices **138**.

(22) Optionally, these media devices and assistant devices associated with the same user account are distributed across different geographical areas. Optionally, these devices are located at the same physical location. Each media or assistant device communicates with another device or the server system **106** using the one or more communication networks **180**. The communication networks **180** used can be one or more networks having one or more type of topologies, including but not limited to the Internet, intranets, local area networks (LANs), cellular networks, Ethernet, Storage Area Networks (SANs), telephone networks, Bluetooth personal area networks (PAN) and the like. In some implementations, two or more devices in a sub-network are coupled via a wired connection, while at least some of devices in the same sub-network are coupled via a local radio communication network (e.g., ZigBee, Z-Wave, Insteon, Bluetooth, Wi-Fi and other radio communication networks).

(23) In various implementations, a first electronic device having a microphone (e.g., a remote control device **104**, any of the media devices **116-126**, assistant device **138**) is coupled to a second electronic device (e.g., a TV device **102** or any of the media devices **116-126**) having one or more processors and memory storing instructions to be executed by the one or more processors via a communication channel. The first electronic device captures an audio signal using its microphone. The audio signal is sampled to a first sequence of audio data samples and a second sequence of audio data samples that follows the first sequence of audio data samples. Optionally, the first and second sequences of audio data samples are recorded in the same recording session or during two distinct recording sessions. Each recording session is activated by a respective user action with the first electronic device (e.g., a user press on an assistant button **606** in FIG. 6). The first electronic device transfers both the first and second sequences of audio data samples to the second electronic device, which then processes each sequence of audio data sample via one of a real time data processing mode and a batch data processing mode.

(24) While or after transferring the first sequence of audio data samples, the second electronic device determines whether communication, computation and storage capabilities of the second electronic device can support processing of audio data samples in the real time data processing mode. If at least one of communication, computation and storage capabilities of the second electronic device cannot support processing of audio data samples in the real time data processing mode, the second electronic device processes the second sequence of audio data samples in the batch data processing mode (e.g., by an audio data processing module **238**, in FIG. 2, which is installed in its operating system or as a user application). In contrast, if the second electronic device can support processing of audio data samples in the real time data processing mode, the second electronic device processes the second sequence of audio data samples in the real time data processing mode (e.g., by an audio manager **234**, in FIG. 2, which is integrated in its operating system). As such, the batch data processing mode is activated in accordance with a determination that the second electronic device cannot support processing of audio data samples in the real time data processing mode, e.g., when an error rate or a latency of audio data exceeds a corresponding error or latency tolerance.

(25) FIG. 2 is an example audio data transfer path **200** between a first electronic device **202** and a second electronic device **204**, in accordance with some implementations. The first electronic device **202** has or is coupled to a microphone **206** configured to capture an audio signal **220**, and the second electronic device **204** is coupled to the first electronic device **202** via a communication channel **208**. The audio signal **220** is sampled at the first electronic device **202**, and audio data samples are transferred from the first electronic device **202** to the second electronic device **204** via the communication channel **208**. The communication channel **208** is enabled by the one or more communication networks **180** including, but not limited to, a local radio communication network (e.g., ZigBee, Z-Wave, Insteon, Bluetooth, Wi-Fi and other radio communication networks). In this example audio data transfer path **200**, the communication channel **208** is formed via a Bluetooth communication link that is enabled collectively by a first Bluetooth stack **208A** of the first electronic device **202** and a second Bluetooth stack **208B** of the second electronic device **204**.

(26) The first electronic device **202** includes an audio streaming module **210** configured to obtain audio data samples of the audio signal **220** captured by the microphone **206** and organize the audio data samples to be transferred over the communication channel **208** to the second electronic device. In some implementations, the audio streaming module **210** groups a subset of the audio data samples into an ordered sequence of audio data packets. Each data packet includes one or more consecutive audio data samples, and optionally has a preamble, message header, encoding packet data, dummy field, and integrity check field in compliance with a predefined data format (e.g., an MPEG-4 HE-AAC codec format, an EVRC voice codec format). At an input to the first Bluetooth stack **208A**, a plurality of ordered sequences of audio data packets are sequentially arranged into a stream of audio data **230** for transmission over the communication channel **208**. In some

implementations, each data packet in the same data packet sequence corresponds to a predefined data format, while two different data packet sequences optionally correspond to the same data format or different data formats.

(27) The second electronic device **204** includes three levels of programs, i.e., a kernel on a hardware abstraction layer (HAL) **216**, device firmware **218**, and application and service programs **222**. The kernel on the HAL **216** and device firmware **218** are part of an operating system of the second electronic device **204**, while the application and service programs **222** are external to the operating system and installed by a manufacturer or user to implement specialized computer operations (e.g., gaming, web browsing, document editing, media playing). In some implementations, after receiving operation data that optionally originate from the first electronic device **202** or any other electronic device via the second Bluetooth stack **208B**, the second electronic device **204** passes the received operation data to an input event identifier **226** of the kernel on the HAL **216**. The input event identifier **226** identifies the operation data from the data received by the second Bluetooth stack **208B**, and provides the identified operation data to an input dispatcher **228** of the device firmware **218**. The input dispatcher **228** assigns the operation data to an assistant application **232** that is installed on the second electronic device **204** to recognize user requests in the operation data and initiate operations in response to the user requests.

(28) In some implementations, the stream of audio data **232** is collected by the microphone **206** of the first electronic device **202** is directly dispatched by the Kernel/HAL **216** to a remote control application **236** associated with the first electronic device **202**. The remote control application **236** collaborates (**260**) with an audio manager **234** of the device firmware **218** and the assistant application **232** installed on the second electronic device **204** to recognize and respond to the user requests in the stream of audio data **230**. The audio manager **234** implements a real time data processing mode and processes audio data **230** on a data packet level. That said, the audio manager **234** is configured to identify the audio data samples from the data packets in the plurality of ordered sequences of audio data packets received from the first electronic device **202**, compensates for erroneous, missing, and out-of-order data packets, organizes the received audio data samples to another sequence of audio data packets, and passes (**262**) the sequence of audio data packets to the assistant application **232** for subsequent audio processing or transfer.

(29) Alternatively, in some implementations, the second electronic device **204** includes a file-based audio data processing module **238** that implements a batch data processing mode and process audio data **230** received from the first electronic device **202** to a data file. The file-based audio data processing module **238** is either integrated (**238A**) in the device firmware **218** (i.e., an operating system) or installed (**238B**) as a user application among the application and service programs **222**. The remote control application **236** provides (**270** or **280**) the audio data **230** to the processing module **238**. The processing module **238** is configured to identify the audio data samples from the data packets in the plurality of ordered sequences of audio data packets received from the first electronic device **202**, caches the data packets in a data file, and provides (**272** or **282**) the data file to the assistant application **232** for subsequent audio processing or transfer. The second electronic device **204** further includes an audio data buffer **214** for storing the data file.

(30) The audio data samples are processed by the audio manager **234** and the processing module **238** alternately in a real time data processing mode and in a batch data processing mode, respectively. The second electronic device **204** switches between these two audio data processing modes depending on whether communication, computation and storage capabilities of the second electronic device **204** can support real time transfer of audio data samples. In some implementations, such determination and mode switching are implemented dynamically, while an ordered sequence of audio data packets is being processed by the audio manager **234** or processing module **238**. Alternatively, such determination and mode switching are implemented between two distinct recording sessions, i.e., between two distinct sequences of audio data packets each of which is independently processed by one of the real time and batch data processing modes. Each

recording session is optionally activated by a user action on the first electronic device **202** or a voice activation detected from the audio data **230** collected by the electronic device **202**.

(31) More specifically, upon receiving the stream of audio data **230**, the second electronic device **204** monitors the audio data **230** to determine whether communication, computation and storage capabilities of the second electronic device **204** can support processing of audio data samples in the real time data processing mode. For example, the second electronic device **204** determines in real time whether a data sample latency of data samples in the audio data **230** exceeds a latency tolerance, e.g., before or after the audio manager **234** processes the data samples in the audio data **230**. In another example, the second electronic device **204** determines, in real time, whether a rate of missing audio data samples of data samples in the audio data **230** exceeds a missing rate tolerance or a rate of out-of-order audio data samples of data samples in the audio data **230** exceeds an out-of-order rate tolerance, e.g., before or after the audio manager **234** processes the data samples in the audio data **230**. In some implementations, the second electronic device **204** monitors its central processing unit (CPU) utilization rate and determines that it cannot support processing of audio data samples in real time if the CPU utilization rate exceeds a predetermined utilization percentage (e.g., 85%). In accordance with the determination result, the second electronic device **204** selects the audio manager **234** or the file-based audio data processing module **238** to process the stream of audio data **230** in real time or in batch, respectively.

(32) In some implementations, the second electronic device **204** is configured to process an output of the audio manager **234** or the processing module **238** locally to recognize the user requests therefrom for the purposes of protecting privacy of a user, and optionally provide the processed output to the remote control application **236** for controlling the first electronic device **202**.

Alternatively, in some implementations, the second electronic device **204** is configured to pre-process the output of the audio manager **234** or the processing module **238** prior to sending it to a remote server system **106** to recognize the user requests therefrom. Alternatively, in some implementations, the second electronic device **204** has a limited speech recognition capability, e.g., when the second electronic device **204** is intended to be a low cost device. The second electronic device **204** is configured to send the entire output of the audio manager **234** or the processing module **238** to the remote server system **106** and rely on the server system **106** to recognize the user requests therefrom.

(33) FIG. 3 is a schematic diagram illustrating an example audio data processing process **300** switching from a real time data processing mode to a batch data processing mode in accordance with some implementations. An audio signal **220** is captured using a microphone **206** of a first electronic device **202**, and sampled at an audio sampling rate to obtain a first sequence of audio data samples **302** and a second sequence of audio data samples **304** that follows the first sequence of audio data samples **302**. The first sequence of audio data samples **302** is transferred to a second electronic device **204** via a communication channel **208**. In some implementations, audio data samples in the first sequence **302** and the second sequence **304** are grouped into a plurality of audio data packets, and each data packet includes one or more consecutive audio data samples. Optionally, the second sequence **304** of audio data samples immediately follows the first sequence **302**. Optionally, the second sequence **304** of audio data samples is separated from the first sequence **302** by a break. The plurality of audio data packets is streamed to the second electronic device **204**. In some situations, one or more data packets are reordered or dropped from the first and second sequences **302** and **304** during the course of being transferred over the communication channel **208**.

(34) Audio data packets **308** are grouped from the first sequence of audio data samples **302**, and processed by the second electronic device **204** in the real time processing mode where erroneous, missing, and out-of-order data packets in the audio data packets **308** may both occur and be corrected. That said, one or more data packets **308** are dropped or reordered by the second electronic device **204** (specifically, by the audio manager **234**). While or after processing the first

sequence of audio data samples **302**, the second electronic device **204** determines that at least one of communication, computation and storage capabilities of the second electronic device **204** cannot support processing of audio data samples in the real time data processing mode. In response to such a determination (at a time $t_{sub.F}$), the second electronic device **204** aborts the real time data processing mode and initiates a batch data processing mode to process the second sequence of audio data samples **304**. Specifically, the second electronic device **204** caches in an audio data buffer **214** the second sequence of audio data samples **304**, and generates a data file **310** including the second sequence of audio data samples **304** in the batch data processing mode. In some implementations, the second electronic device **204** determines a corresponding capability of supporting the processing of audio data samples in the real time data processing mode based on at least one of: a data sample latency, a rate of missing audio data samples, a rate of out-of-order audio data samples, and a CPU utilization rate associated with the second electronic device **204**.

(35) In some implementations, the second electronic device **204** ceases processing of the second sequence of audio data samples **304** in accordance with a determination that communication, computation and storage capabilities of the second electronic device **204** can support processing of audio data samples in the real time data processing mode. Alternatively, in some implementations, the second electronic device **204** is configured to limit the second sequence **304** to include a predefined number of audio data samples. The second electronic device **204** processes the second sequence of audio data samples **304** to be cached in a first data file **310** when the predefined number is reached. The second electronic device **204** then organizes a third sequence of audio data samples **306** that immediately follows the second sequence of audio data samples **304** in a second data file **312**. The second data file **312** is transferred when the predefined number of audio data samples are included in the third sequence of audio data samples **306** or when it is determined that the second electronic device **204** can support processing of audio data samples in the real time data processing mode.

(36) Under some circumstances, a current number of audio data samples included in a data file **310** or **312** has not reached the predefined number when it is determined that the second electronic device **204** can support processing of audio data samples in the real time data processing mode. In accordance with the determination, the data file **310** or **312** may be transferred immediately with the current number of audio data samples. Optionally, the data file **310** or **312** is aborted from being transferred, and the current number of audio data samples are re-organized into data packets for real time audio data processing by the audio manager **234** of the second electronic device **204**.

(37) In some implementations, the processed first sequence of audio data samples **308** and the data file **310** including the second sequence of audio data samples are transferred (**320**) to a server system **106**. The processed first sequence of audio data samples **302** has a first data transfer rate corresponding to the real time data processing mode and the second sequence of audio data samples **304** has a second data transfer rate corresponding to the batch data processing mode. The second data transfer rate is greater than the first data transfer rate. In some implementations, the first data transfer rate is slower than the audio sampling rate of the audio signal **220**, and the second data transfer rate is greater than the audio sampling rate.

(38) FIG. **4** is a schematic diagram illustrating an example audio data processing process **400** switching from a batch data processing mode to a real time data processing mode in accordance with some implementations. An audio signal **220** is captured using a microphone of a first electronic device **202**, and sampled at an audio sampling rate to obtain a first sequence of audio data samples **402** and a second sequence of audio data samples **404** that follows the first sequence of audio data samples **402**. Both the first sequence of audio data samples **402** and the second sequence of audio data samples **404** are transferred to a second electronic device **204** via a communication channel **208**. The first sequence of audio data samples **302** is processed by the second electronic device **204** according to a batch data processing mode. Audio data samples in the first sequence **402** are cached in an audio data buffer **214** of the first electronic device **202** and

organized into a data file **406**.

(39) While or after processing the first sequence of audio data samples **402**, the second electronic device **204** detects or determines that the second electronic device **204** can support processing of audio data samples in a real time data processing mode. Optionally, in accordance with such a determination (at a first time t.sub.A), the second electronic device **204** continues to add more audio data samples to the first sequence **402** until a number of the audio data samples of the first sequence **402** reaches a predefined number at a second time t.sub.B. The second electronic device **204** completes caching the first sequence of audio data samples **402** in the data file **406**, before it starts to process, in the real time data processing mode, the second sequence of audio data samples **404** collected subsequently to the first sequence of audio data samples **402**. Alternatively, in some implementations, in accordance with the determination (at a time t.sub.C), the second electronic device **204** ceases adding audio data samples to the first sequence **402** independently of whether a number of audio data samples to the first sequence **402** has reached the predefined number. The second electronic device **204** prepares the data file **406** at the time t.sub.C, thereby terminating the batch data processing mode. The second electronic device **204** immediately starts to transfer the second sequence of audio data samples **404** collected subsequently to the first sequence of audio data samples **402** in the real time data processing mode. Further, in some implementations (not shown), upon the determination (e.g., at the time t.sub.C), the second electronic device **204** aborts processing the first sequence **402** of audio data samples in the batch data processing mode and starts to process the first sequence of audio data samples **402** into data packets **408** immediately in the real time data processing mode. After the first sequence of audio data samples **402** is transferred, the second electronic device **204** continues to process the second sequence of audio data samples **404** in the real time data processing mode.

(40) Each of the first sequences of audio data samples **302** and **402** optionally begins the stream of audio data **230** sent to the second electronic device **204** or is in the middle of the stream of audio data **230**. Likewise, each of the second sequences of audio data samples **304** and **404** and the third sequence of audio data samples **306** optionally is the last sequence in the stream of audio data **230** sent to the second electronic device **204** or is in the middle of the stream of audio data **230**. It is noted that in some implementations, the first sequence of audio data samples **302** or **402** does not immediately precedes the second sequence of audio data samples **304** or **404**. The first and second sequences are captured during two distinct recording sessions separated by a break. The second electronic device **204** determines whether it can support processing of audio data samples in the real time data processing mode during the break separating the two recording sessions.

(41) FIG. 5 is a schematic diagram illustrating an example voice assistant process **500** initiated by a user action or a voice input in accordance with some implementations. The voice assistant process **500** is collaboratively implemented by a first electronic device **202** and a second electronic device **204**. In some implementations, the first electronic device **202** includes a physical assistant button (e.g., an assistant button **606** in FIG. 6), and a user is allowed to request initiation or termination of voice assistant functions via a user action on the assistant button. For example, the user applies a press on the physical assistant button to initiate the voice assistant process **500** (also called a recording session). While the user holds the press on the assistant button, an audio signal is continuously collected by a microphone **206** of the first electronic device **202** for audio data sampling, transfer and recognition via the voice assistant process **500**. The microphone **206** does not stop collecting the audio signal, until the user releases the press on the assistant button to complete the corresponding recording session.

(42) In some implementations, in response to detection of a first user action, the first electronic device **202** sends an assistant invocation request **502** to a remote control application **236** and an assistant application **232** of the second electronic device **204**. In response to the assistant invocation request **502**, the assistant application **232** verifies that the first electronic device **202** is permitted to implement the voice assistant process **500** with the second electronic device **204**, and sends an

instruction to start assistant **504** to the remote control application **236**. In response to the instruction to start assistant **504**, the remote control application **236** sends an open microphone instruction **506** to the first electronic device **202**. After the microphone **206** of the first electronic device **202** is turned on, audio data samples **230** are collected and transferred to the second electronic device **204**. After being transferred to the second electronic device **204**, the audio data samples **230** are processed (**520** and **530**) by the audio manager **234** associated with a real time data processing mode or by the file-based audio data processing module **238** associated with a batch data processing mode. In some implementations, the real time and batch data processing modes alternate dynamically at the second electronic device **204**, e.g., based on a data sample latency, rate of missing audio data samples, rate of out of order audio data samples, or CPU utilization rate associated with the second electronic device **204**. In some implementations, only one of the real time and batch data processing modes is activated during each recording section activated in response to detection of the first user action.

(43) In some implementations, the assistant application **232** sends an instruction to start recording **508** to the remote control application **236**, such that the remote control application **236** may control the first electronic device **202** to capture the audio data collected by the first electronic device **202**. The instruction to start recording **508** is optionally issued with the instruction to start assistant **504** and configured to trigger the open microphone instruction **506**. In response to the instruction to start recording **508**, the audio data samples **230** are recorded by the second electronic device **204** after they are transferred from the first electronic device **202**. Alternatively, in some implementations, the instruction to start recording **508'** is issued by the assistant application after a subset of the audio data samples **230** have been received by the second electronic device **204**. The instruction to start recording **508'** may be issued based on content of the subset of the audio data samples (e.g., a user request in the content), and the subset of the audio data samples **230** is not recorded by the second electronic device **204**.

(44) In some implementations, the assistant application **232** sends an instruction to stop recording **510** to both an audio manager **234** and the remote control application **236**, such that the remote control application **236** may issue a close microphone instruction **512** to control the first electronic device **202** to turn off its microphone **206**. Alternatively, in some implementations, the user of the first electronic device **202** terminates the first user action that initiated the voice assistant process **500** or applies a second user action (e.g., a release of the assistant button) to terminate the voice assistant process **500**. In response to the second user action, the microphone **206** of the first electronic device **202** is turned off, and a request for ending assistant **514** is sent to the remote control application **236** and assistant application **232** of the second electronic device **204**.

(45) Referring to FIGS. **3** and **4**, the microphone **206** of the first electronic device **202** is set to be turned on and start to capture the audio signal at different times with respect to the first user press. In some implementations, the first electronic device **202** receives a first user action requesting recording of the audio signal at a first time $t_{sub.1}$, and its microphone **206** is turned on to capture the audio signal immediately in response to the first user action at a second time $t_{sub.2}$, independently of whether the open microphone instruction **506** is issued from the second electronic device **204**. The second time $t_{sub.2}$ is substantially concurrently with the first time $t_{sub.1}$. Alternatively, in some implementations, the first electronic device **202** receives a first user action requesting recording of the audio signal at a first time $t_{sub.1'}$. The audio signal is captured by the microphone at a second time $t_{sub.2}$ that precedes the first time $t_{sub.1'}$. The audio data corresponding to a duration between the times $t_{sub.2}$ and $t_{sub.1'}$ are cached, but still transferred to the second electronic device **204** in response to the first user action. Additionally, in some implementations, the first electronic device **202** receives a first user action requesting recording of the audio signal at a first time $t_{sub.1''}$, and starts to capture the audio signal at a second time $t_{sub.2}$ that is subsequent to the first time $t_{sub.1''}$, e.g., delayed from the first time $t_{sub.1''}$ by a predefined buffer time (such as 5 seconds).

(46) It is noted that in some implementations, the first electronic device **202** waits to receive an audio data request including the open microphone instruction **506** from the second electronic device **204**, in a duration of time between receiving the first user action time ($t_{sub.1}$, $t_{sub.1'}$, or $t_{sub.1''}$) and transferring the captured audio data to the second electronic device **204** at $t_{sub.2}$. The second electronic device **204** obtains an approval to send the audio data request in response to the first user action, and this approval is granted by the assistant application **232** of the second electronic device **204** or by a remote server system **106**.

(47) In some situations, the voice assistant process **500** is initiated by a voice input. The microphone **206** of the first electronic device **202** is configured to continuously collect audio signals and provide corresponding audio data to the second electronic device **204** regardless of whether the first electronic device **202** is at a sleep mode and an active mode. At the sleep mode, the audio data is not processed to recognize any user requests for the voice assistant functions that are used to control media devices or user applications, until one or more predefined hot words (e.g., "Hi Google") are detected to enable the active mode. The second electronic device **204** is configured to detect one or more predefined hot words in the audio data locally and initiates the voice assistant process **500** in response to detection of the one or more predefined hot words.

(48) Referring to FIG. 5, in some implementations, the assistant invocation request **502** includes one or more predefined hot words. The second electronic device **204** confirms the assistant invocation request **502** upon detecting the hot words in the audio data received from the first electronic device **202**. In response to the assistant invocation request **502**, the assistant application **232** verifies that the first electronic device **202** is permitted to implement the voice assistant process **500** with the second electronic device **204**, and sends an instruction to start assistant **504** to the remote control application **236**. The second electronic device **204** is controlled to operate in an active mode to process the audio data samples **230** locally in a real time data processing mode or a batch data processing mode to identify additional user requests. Further, in some implementations, the assistant application **232** identifies a user request to stop recording (e.g., "Google, stop recording" in FIGS. 3 and 4) in the audio data **230** provided by the first electronic device. The assistant application **232** generates an instruction to stop recording **510** in the audio data samples **230**, and send the instruction **510** to an audio manager **234** and the remote control application **236**, such that the remote control application **236** may issue a close microphone instruction **512** to control the first electronic device **202** to turn off its microphone. By these means, the voice assistant process **500** is initiated and terminated base on the audio data captured by the first electronic device **202** without using any physical user action.

(49) Referring to FIG. 3, in an example, a request **314** to cease capturing the audio signal is recognized in the audio signal **220**, and sent from the second electronic device **204** to the first electronic device **202**. In response to the request **314**, the second electronic device **204** completes processing of the first data file **310** including the second sequence of data samples **304**, but aborts transferring the second data file **312** including the third sequence of data samples **306** that immediately follows the second sequence **304**. In contrast, referring to FIG. 4, in another example, upon receiving the request **314**, the second electronic device **204** aborts processing subsequent data packets that follow the request **314** in the second sequence of audio data samples **404**.

(50) FIG. 6 illustrates an example remote control device **104** configured to transfer audio data to a television device, in accordance with some implementations. The plurality of user buttons of the remote control device **104** includes one or more of a power button **602**, a home button **604**, an assistant button **606**, a cycle button **608** (also called a play/cycle button), a previous button **610**, a next button **612**, a preview/backdrop button **614** and a volume control button **616**. A user actuation on the power button **602** powers on a media device coupled to the remote control device **104** when the media device is off prior to the user actuation, and powers off the media device when the media device is on prior to the user actuation. A user actuation on the home button **604** controls the media device coupled to the remote control device **104** to display a home screen. For example, the home

screen displays a specific advertisement clip or a randomly selected media program provided by a predetermined Internet content channel. In some implementations, the power button **602** or the home button **604** functions as a quick cast button configured to enable immediate play of media content provided by a specific Internet content channel.

(51) A user action on the assistant button **606** controls a microphone **206** integrated in the remote control device **104** to collect audio signals in the media environment **100**, and user requests are extracted from the audio signals to control one or more media play devices (e.g., the TV device **102**) located in the media environment **100**. In some implementations, the microphone **206** of the remote control device **104** initiates collecting audio signals from the environment **100** when a first brief press is applied on the assistant button **606**, and a second brief press or a user request is applied to cease collecting the audio signals. Alternatively, in some implementations, the microphone **206** of the remote control device **104** only collects audio signals from the environment **100** when the assistant button **606** is being pressed, and ceases collecting the audio signals when the assistant button **606** is released. Additionally, in some implementations, the microphone **206** of the remote control device **104** continuously captures audio signals from the environment **100**, and the audio signals include one or more predefined hot words and/or user requests. The user requests can be used to control the remote control device **104** or one or more media devices or applications coupled to the remote control device **104**.

(52) In this application, audio data is processed at the second electronic device **204** (e.g., a network-connected television device **102**) having two audio data processing modes that includes a real-time data processing mode and a batch data processing mode. The second electronic device **204** determines whether communication, caching and processing capabilities of the second electronic device **204** can support processing of audio data samples in the real time data processing mode. In accordance with a determination that the second electronic device **204** can support processing of audio data samples in the real time data processing mode, the second electronic device **204** processes subsequent audio data samples according to the real time data processing mode on a data packet level. In accordance with a determination that the second electronic device **204** cannot support processing of audio data samples in the real time data processing mode, the second electronic device **204** caches, in a buffer of the second electronic device **204**, subsequent audio data samples, and generates a data file including these audio data samples in a batch data processing mode. In some implementations, determination and mode enabling are implemented during a break between two recording sessions. Alternatively, in some implementations, determination and mode switching are implemented dynamically during the same recording session.

(53) FIG. 7 is a flow chart of a method **700** for processing audio data dynamically in two audio data processing modes (from a real time data processing mode to a batch data processing mode), in accordance with some implementations. The method **700** is performed a first electronic device **202** and a second electronic device **204**, and optionally governed by instructions that are stored in a non-transitory computer readable storage medium and that are executed by one or more processors of the respective electronic device. Each of the operations shown in FIG. 7 may correspond to instructions stored in a computer memory or non-transitory computer readable storage medium. The computer readable storage medium may include a magnetic or optical disk storage device, solid state storage devices such as Flash memory, or other non-volatile memory device or devices. The instructions stored on the computer readable storage medium may include one or more of: source code, assembly language code, object code, or other instruction format that is interpreted by one or more processors. Some operations in the method **700** may be combined and/or the order of some operations may be changed.

(54) Audio signals are captured (**702**) using a microphone **206** of the first electronic device **202**. The first electronic device **202** obtains (**704**) from the audio signal a first sequence of audio data samples **302** and a second sequence of audio data samples **304** that follows the first sequence of

audio data samples **302**, and transfers to the second electronic device **204** via a communication channel **208** the first sequence of audio data samples **302** according to a real time data processing mode. The second electronic device **204** receives (**706**), from the first electronic device **202** via the communication channel **208**, the first sequence of audio data samples **302** and the second sequence of audio data samples **304**. While or after the first sequence of audio data samples **302**, the second electronic device **204** determines (**708**) that the second electronic device **204** cannot support processing of audio data samples in the real time data processing mode. In accordance with the determination that the second electronic device **204** cannot support processing of audio data samples in the real time data processing mode, the second electronic device **204** caches (**712**), in a buffer of the second electronic device, the second sequence of audio data samples **304**, and generates (**714**) a data file **310** including the second sequence of audio data samples **304** in the batch data processing mode.

(55) In some implementations, the second electronic device **204** transfers (e.g., **540** in FIG. 5) the processed first sequence of audio data samples and the data file including the second sequence of audio data samples to a server system **106**. The first sequence of audio data samples **302** has a first data transfer rate correspond to the real time data processing mode, and the second sequence of audio data samples **304** has a second data transfer rate corresponding to the batch data processing mode. The second data transfer rate is greater than the first data transfer rate. Further, in an example, the audio signals captured by the first electronic device **202** are sampled at an audio sampling rate to obtain the first and second sequences of audio data samples **302** and **304**. The first data transfer rate is slower than the audio sampling rate, and the second data transfer rate is greater than the audio sampling rate.

(56) In some implementations, the audio data samples in the first sequence **302** are grouped into a plurality of audio data packets in the real time data processing mode. Each audio data packet includes one or more consecutive audio data samples that are optionally organized according to a consistent data format. The plurality of audio data packets are streamed to a server system **106**.

(57) In some implementations, the second electronic device **204** is determined as not supporting the transferring of audio data samples in the real time data processing mode based on at least one of: a data sample latency, a rate of missing audio data samples, and a rate of out-of-order audio data samples associated with the second electronic device **204**. Specifically, in an example, the data sample latency of a subset of the processed first sequence of audio data samples exceeds a latency tolerance. In another example, a rate of missing audio data samples of the processed first sequence of audio data samples **302** exceeds a missing rate tolerance. In yet another example, a rate of out-of-order audio data samples of the processed first sequence of audio data samples **302** exceeds an out-of-order rate tolerance.

(58) In some implementations, a first user action requesting recording of the audio signals are received at the first electronic device **202**. The audio signals are captured in response to the first user action. An example is the first user action is a press on an assistant button of the first electronic device **202**. The press initiate a process to obtain an approval by an assistant application **232** of the second electronic device **204** or a server system **106**. Upon receiving the approval, the audio signals are being captured, processed, and recorded. Specifically, in an example, the first electronic device **202** receives the first user action requesting recording of the audio signals at a first time $t_{sub.1}$, wherein capturing the audio signals is initiated at a second time $t_{sub.2}$ subsequent to the first time $t_{sub.1}$. The second time $t_{sub.2}$ is delayed from the first time by a predefined buffer time. Specifically, in some situations, in response to the first user action, the first electronic device **202** receives an audio data request from the second electronic device **204**. The second electronic device **204** is configured to obtain an approval to send the audio data request in response to the first user action. Transferring the first sequence of audio data samples is initiated in response to the audio data request.

(59) In some implementations, the data file **310** includes a first data file **310**. After generating the

first data file **310**, the second electronic device **204** continues to generate a second data file **312** including a third sequence of audio data samples **306** in the batch data processing mode. The third sequence of audio data samples **306** immediately follows the second sequence of audio data samples **304** in the audio signals, and each of the second and third sequences of audio data samples **304** and **306** has a predefined number of data samples.

(60) In some implementations, the second electronic device **204** is configured to transfer the processed first and second sequences of audio data samples **302** and **304** to a server system **106** for audio processing (e.g., voice recognition). The server system **106** hosts a virtual user domain including a user account. The first and second electronic devices **202** and **204** are linked to the user account.

(61) Alternatively, in some implementations, the audio signals include one or more predefined hot words or a user request. The second electronic device **204** is configured to process the first and second sequences of audio data samples locally to identify the one or more predefined hot words or user request in the audio signal. Further, in some implementations, the user request includes a request to cease capturing the audio signal. The request to cease capturing the audio signals are recognized by the second electronic device **204** and provided to the first electronic device **202** thereby. In response to the request, the first electronic device **202** aborts transferring the sequence of audio data samples that follows the second sequence of audio data samples **304**.

(62) In some implementations, while transferring the second sequence of data samples **304**, the second electronic device **204** receives a second user action for ceasing capturing the audio signal. In response to the second user action, the second electronic device **204** aborts receiving a sequence of audio data samples that immediately follows the second sequence of audio data samples **304**.

(63) FIG. **8** is a flow chart of a method **800** for processing audio data dynamically in two audio data processing modes (from a batch data processing mode to a real time data processing mode), in accordance with some implementations. The method **800** is performed by a first electronic device **202** and a second electronic device **204**, and optionally governed by instructions that are stored in a non-transitory computer readable storage medium and that are executed by one or more processors of the respective electronic device. Each of the operations shown in FIG. **8** may correspond to instructions stored in a computer memory or non-transitory computer readable storage medium. The computer readable storage medium may include a magnetic or optical disk storage device, solid state storage devices such as Flash memory, or other non-volatile memory device or devices. The instructions stored on the computer readable storage medium may include one or more of: source code, assembly language code, object code, or other instruction format that is interpreted by one or more processors. Some operations in the method **800** may be combined and/or the order of some operations may be changed.

(64) Audio signals are captured (**802**) using a microphone of the first electronic device **202**. The first electronic device **202** obtains (**804**) a first sequence of audio data samples **402** and a second sequence of audio data samples **404** that follows the second sequence of audio data samples **402** in the audio signals. The second electronic device **204** receives (**806**), from a first electronic device **202**, the first sequence of audio data samples **402** and the second sequence of audio data samples **404**. The second electronic device **204** processes the first sequence of audio data samples **402** according to the batch data processing mode, including caching (**810**), in a buffer of a second electronic device, the first sequence of audio data samples **402**, and generating (**812**) a data file including the first sequence of audio data samples **402**. While or after processing the first sequence of audio data samples, the second electronic device **204** determines (**814**) that the second electronic device **204** can support processing of audio data samples in a real time data processing mode. In accordance with the determination that the second electronic device **204** can support processing of audio data samples in the real time data processing mode, the second electronic device **204** processes (**816**) the second sequence of audio data samples according to the real time data processing mode.

(65) In some implementations, the audio data samples **402** and **404** are transferred to a server system **106**. The first sequence of audio data samples **402** has a first data transfer rate corresponding to the real time data processing mode, and the second sequence of audio data samples **404** has a second data transfer rate corresponding to the batch data processing mode. The first data transfer rate is greater than the second data transfer rate. Further, in an example, the second data transfer rate is slower than the audio sampling rate, and the first data transfer rate is greater than the audio sampling rate. In some implementations, the audio data samples in the second sequence **404** are grouped into a plurality of audio data packets in the real time data processing mode, and the plurality of audio data packets are optionally streamed to the server system **106** in real time.

(66) Referring to FIGS. **7** and **8**, in some implementations, the first electronic device **202** includes a remote control device **104**, and the second electronic device **204** includes a network-connected TV device **102** configured to be controlled by the remote control device **104**. In some implementations, the second electronic device **204** includes one or more processors, and memory storing one or more programs configured to implement an Android operating system and one or more user applications on the second electronic device **204**. In some implementations, the first electronic device **202** is battery powered.

(67) It should be understood that the particular order in which the operations in each of FIGS. **7** and **8** have been described are merely exemplary and are not intended to indicate that the described order is the only order in which the operations could be performed. One of ordinary skill in the art would recognize various ways to display information items and focus content in a unified user interface as described herein. Additionally, it should be noted that details described with respect to one of the methods **700**, **750**, **800** and **850** are also applicable in an analogous manner to any other ones of the methods **700**, **750**, **800** and **850**. For brevity, the analogous details are not repeated.

(68) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first electronic device can be termed a second electronic device, and, similarly, a second electronic device can be termed a first electronic device, without departing from the scope of the various described implementations. The first electronic device and the second electronic device are both electronic device, but they are not the same electronic device.

(69) The terminology used in the description of the various described implementations herein is for the purpose of describing particular implementations only and is not intended to be limiting. As used in the description of the various described implementations and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(70) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting” or “in accordance with a determination that,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event]” or “in accordance with a determination that [a stated condition or event] is detected,” depending on the context.

(71) Although various drawings illustrate a number of logical stages in a particular order, stages that are not order dependent may be reordered and other stages may be combined or broken out.

While some reordering or other groupings are specifically mentioned, others will be obvious to those of ordinary skill in the art, so the ordering and groupings presented herein are not an exhaustive list of alternatives. Moreover, it should be recognized that the stages can be implemented in hardware, firmware, software or any combination thereof.

(72) The foregoing description, for purpose of explanation, has been described with reference to specific implementations. However, the illustrative discussions above are not intended to be exhaustive or to limit the scope of the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The implementations were chosen in order to best explain the principles underlying the claims and their practical applications, to thereby enable others skilled in the art to best use the implementations with various modifications as are suited to the particular uses contemplated.

Claims

1. A method of processing audio data, comprising: receiving, by a second electronic device, a first sequence of audio data samples and a second sequence of audio data samples from a first electronic device, wherein the second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device; processing, by the second electronic device, the first sequence of audio data samples according to a real time data processing mode; determining that the second electronic device cannot support processing of audio data samples in the real time data processing mode; and in accordance with the determination that the second electronic device cannot support processing of audio data samples in the real time data processing mode, caching, in a buffer of the second electronic device, the second sequence of audio data samples, and generating a data file including the second sequence of audio data samples in a batch data processing mode.
2. The method of claim 1, further comprising: transferring the processed first sequence of audio data samples and the data file including the second sequence of audio data samples to a server system; wherein the first sequence of audio data samples has a first data transfer rate corresponding to the real time data processing mode, and the second sequence of audio data samples has a second data transfer rate corresponding to the batch data processing mode; and wherein the second data transfer rate is greater than the first data transfer rate.
3. The method of claim 2, wherein: the audio signals are sampled at an audio sampling rate to obtain the first and second sequences of audio data samples; and the first data transfer rate is slower than the audio sampling rate, and the second data transfer rate is greater than the audio sampling rate.
4. The method of claim 1, wherein the method further comprises: grouping the audio data samples in the first sequence into a plurality of audio data packets, each audio data packet including one or more consecutive audio data samples; and streaming the plurality of audio data packets to a server system.
5. The method of claim 1, wherein determining that the second electronic device cannot support the processing of audio data samples in the real time data processing mode further comprises at least one of: determining that a data sample latency of a subset of the processed first sequence of audio data samples exceeds a latency tolerance; determining that a rate of missing audio data samples of the processed first sequence of audio data samples exceeds a missing rate tolerance; and determining that a rate of out-of-order audio data samples of the processed first sequence of audio data samples exceeds an out-of-order rate tolerance.
6. The method of claim 1, wherein the second electronic device is determined as not supporting the transferring of audio data samples in the real time data processing mode based on at least one of: a data sample latency, a rate of missing audio data samples, or a rate of out-of-order audio data samples associated with the second electronic device.

7. The method of claim 1, further comprising: receiving, by the first electronic device, a first user action requesting recording of the audio signals, wherein the audio signals are captured in response to the first user action.
8. The method of claim 1, further comprising: receiving, by the first electronic device, a first user action requesting recording of the audio signals at a first time, wherein capturing the audio signals is initiated at a second time subsequent to the first time.
9. The method of claim 8, further comprising: in response to the first user action, obtaining an approval to send an audio data request and generating an audio data request by the second electronic device; and wherein receiving the first sequence of audio data samples is in response to the audio data request.
10. The method of claim 1, wherein the data file includes a first data file, and wherein the method further comprises: after generating the first data file, continuing to generate, by the second electronic device, a second data file including a third sequence of audio data samples in the batch data processing mode, wherein the third sequence of audio data samples immediately follows the second sequence of audio data samples in the audio signals, and each of the second and third sequences of audio data samples has a predefined number of data samples.
11. The method of claim 1, wherein: the second electronic device is configured to transfer the processed first and second sequences of audio data samples to a server system for audio processing; the server system hosts a virtual user domain including a user account; and the first and second electronic devices are linked to the user account.
12. The method of claim 1, wherein the audio signals include one or more predefined hot words or a user request, and the second electronic device is configured to process the first and second sequences of audio data samples locally to identify the one or more predefined hot words or user request in the audio signals.
13. The method of claim 12, wherein the user request includes a request to cease capturing the audio signals, and wherein the method further comprises: generating a request to cease capturing the audio signals by the second electronic device, the first electronic device configured to in response to the request, abort transferring a sequence of audio data samples that follows the second sequence of audio data samples.
14. The method of claim 1, further comprising: while transferring the second sequence of data samples, receiving a second user action for ceasing capturing the audio signals; and in response to the second user action, aborting receiving a sequence of audio data samples that immediately follows the second sequence of audio data samples.
15. The method of claim 1, wherein the first electronic device includes a remote control device, and the second electronic device includes a network-connected television device configured to be controlled by the remote control device.
16. The method of claim 1, wherein the second electronic device includes one or more processors, and memory storing one or more programs configured to implement an Android operating system and one or more user applications on the second electronic device.
17. The method of claim 16, wherein the Android operating system includes an audio manager module having instructions for processing the first sequence of audio data samples according to the real time data processing mode.
18. The method of claim 16, wherein the one or more user applications of the second electronic device includes a file-based audio data processing module for caching the second sequence of audio data samples in the buffer and generating the data file including the second sequence of audio data samples in the batch data processing mode.
19. The method of claim 1, wherein the second electronic device is determined as not support processing of audio data samples in the real time data processing mode during a recording session activated by a user action, and both the first and second sequences of data samples are processed during the recording session.

20. The method of claim 1, wherein the second electronic device is determined as not support processing of audio data samples in the real time data processing mode in a break separating two distinct recording sessions that are activated by two distinct user actions, and the first and second sequences of data samples are captured during the two distinct recording sessions.

21. A method of processing audio data, comprising: receiving, by a second electronic device, a first sequence of audio data samples and a second sequence of audio data samples from a first electronic device, wherein the second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device; processing, by the second electronic device, the first sequence of audio data samples according to a batch data processing mode, including caching, in a buffer of a second electronic device, the first sequence of audio data samples, and generating a data file including the first sequence of audio data samples; determining that the second electronic device can support processing of audio data samples in a real time data processing mode; and in accordance with the determination that the second electronic device can support processing of audio data samples in the real time data processing mode, processing by the second electronic device the second sequence of audio data samples according to the real time data processing mode.

22. An electronic device, comprising: one or more processors; and memory having instructions stored thereon, which when executed by the one or more processors cause the processors to perform a method of processing audio data, wherein the one or more processors are configured to: receive, by a second electronic device, a first sequence of audio data samples and a second sequence of audio data samples from a first electronic device, wherein the second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device; process, by the second electronic device, the first sequence of audio data samples according to a real time data processing mode; determine that the second electronic device cannot support processing of audio data samples in the real time data processing mode; and in accordance with the determination that the second electronic device cannot support processing of audio data samples in the real time data processing mode, cache, in a buffer of the second electronic device, the second sequence of audio data samples, and generate a data file including the second sequence of audio data samples in a batch data processing mode.

23. A non-transitory computer-readable medium, having instructions stored thereon, which when executed by one or more processors cause the processors to perform a method of processing audio data, the method comprising: receiving, by a second electronic device, a first sequence of audio data samples and a second sequence of audio data samples from a first electronic device, wherein the second sequence of audio data samples follows the first sequence of audio data samples in audio signals captured by a microphone of the first electronic device; processing, by the second electronic device, the first sequence of audio data samples according to a real time data processing mode; determining that the second electronic device cannot support processing of audio data samples in the real time data processing mode; and in accordance with the determination that the second electronic device cannot support processing of audio data samples in the real time data processing mode, caching, in a buffer of the second electronic device, the second sequence of audio data samples, and generating a data file including the second sequence of audio data samples in a batch data processing mode.
